

Balloon Pop Game

Week 2 Progress Report

Project Name: Balloon Pop Game

Week: 2

Development Tool: Apache NetBeans IDE 28

Programming Language: Java

GUI Library: Java Swing

Objective:

The objective of Week 2 was to develop the basic graphical user interface (GUI) for the Balloon Pop Game using Java Swing. This GUI will serve as the foundation for implementing game functionality in the upcoming weeks.

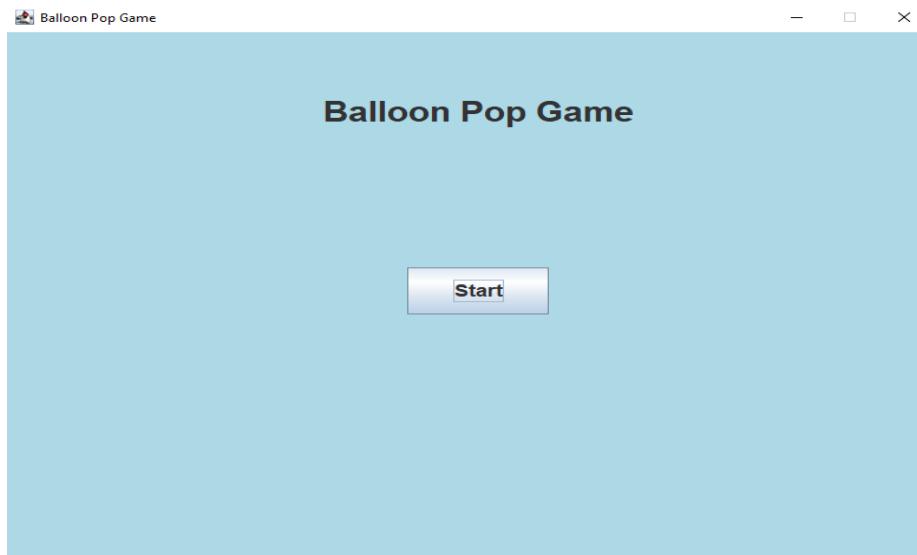
Work Completed:

The following tasks were successfully completed during Week 2:

- 1.Created the main game window using the "JFrame" class.
- 2.Set the window title to "Balloon Pop Game".
- 3.Set the window size to 800×600 pixels.
- 4.Positioned the window at the center of the screen.
- 5.Configured the application to close properly when the close button is clicked.
- 6.Applied a light sky-blue background color.
- 7.Disabled the default layout manager to manually position GUI components.
- 8.Created and added a centered game title using "JLabel".
- 9.Customized the title using the Arial font with bold style and size 30.
- 10.Created a Start button using "JButton".

- 11.Positioned the Start button near the center of the window.
- 12.Customized the Start button using the Arial font with bold style and size 18
- 13.Added both the title label and the Start button to the game window.
- 14.Disabled window resizing to maintain a fixed interface.
- 15.Successfully displayed the completed GUI window.

Screenshot of the GUI :



Output:

The application runs successfully and displays:

- 1.A game window titled Balloon Pop Game.
- 2.A centered game title.
- 3.A centered Start button.
- 4.A light sky-blue background.
- 5.A fixed-size window measuring 800×600 pixels.

The Start button is currently displayed as part of the interface. Its functionality will be implemented in the following weeks according to the project schedule.

Challenges Faced:

During this week, I encountered the following challenges:

- 1.Learning the basics of Java Swing GUI development.
- 2.Understanding the use of "JFrame", "JLabel", and "JButton".
- 3.Learning how to manually position GUI components using "setBounds()".
- 4.Understanding the role of constructors in initializing a Java application.

These challenges were overcome through study, experimentation, and repeated testing.

Learning Outcomes:

At the end of Week 2, I learned:

- 1.How to create a graphical user interface using Java Swing.
- 2.How to create and configure a "JFrame".
- 3.How to use "JLabel" and "JButton".
- 4.How to customize fonts, colors, and window properties.
- 5.How to manually position components inside a window.
- 6.How constructors initialize GUI components.
- 7.The importance of organizing GUI code for future game development.

Conclusion:

Week 2 was completed successfully. A functional and well-organized graphical user interface was developed using Java Swing. This interface provides a solid foundation for implementing gameplay features such as moving balloons, scoring, timers, and user interaction in the upcoming weeks.